

Retrieval from Zero

A beginner's guide to retrieval, reranking, evaluation, and end-to-end search pipelines

Start with the foundations of text, vectors, similarity, and ranking. Then build toward TF-IDF, BM25, embeddings, bi-encoders, cross-encoders, category-aware reranking, and reliable evaluation.

A self-contained study path with worked examples, diagrams, glossaries, and self-checks.

Author: Soroosh Aghaei

How to Use This Guide

This document is designed for a student who has no stable base yet. It is not a compressed revision sheet. It is a teaching document. Read it in order on the first pass. On the second pass, pause often and solve the self-checks before reading the answers. On the third pass, connect the ideas to the actual notebook and its implementation choices.

There are three important promises behind the structure.

- **Nothing important is assumed.** Words such as feature, vector, relevance, candidate set, reranker, or leakage are taught before they are used heavily.
- **Formulas come late.** The guide first answers the question “what problem does this solve?” and only then gives symbols.
- **The notebook comes after the concepts.** The goal is to understand the system, not to memorize variable names blindly.

If you feel lost, remember this

Each chapter tries to follow the same learning rhythm: why the idea exists, what problem it solves, a toy example, a more technical explanation, a careful formal version, a warning about common confusion, and a few small questions so you can check whether the idea really made sense.

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Part I

Before Retrieval: the beginner foundations

What data science is, in plain language

Why this chapter exists

Before learning retrieval, you need a small mental toolkit for thinking about data-driven systems. Without that toolkit, later words such as feature, label, model, training, and evaluation feel abstract and disconnected.

Intuition in ordinary language

Data science is the practice of learning something useful from organized observations. The observations may come from customers, machines, experiments, documents, clicks, medical records, or forum posts. A data scientist tries to answer questions such as: what pattern is present, what can be predicted, what items look similar, or which option should be ranked first.

A good first picture is this: you collect many examples, you choose what information from each example to look at, you decide what outcome matters, and you build a system that uses the past examples to behave better on new ones.

Practical intuition

If a teacher has many old exam papers and their grades, the papers are **data**. The characteristics she notices, such as number of solved questions or number of conceptual mistakes, are **features**. The final grade is the **label**. A system that learns how these characteristics connect to the grade is a **model**.

Concrete toy example

Imagine a tiny email problem. You want to decide whether an email is spam.

- Email A contains “win money now” and many links.
- Email B contains “meeting moved to 3pm” and no suspicious phrases.

You might choose these features: number of links, whether the phrase “win money” appears,

whether the sender is known, and email length. The label is one word: **spam** or **not spam**.

A rule-based system might say: “if the email contains the phrase ‘win money’, call it spam.” A learned system instead looks at many examples and discovers which combinations of features usually lead to spam.

Technical explanation

Here are the beginner definitions that matter most.

- **Data:** recorded examples.
- **Feature:** a measurable description of one example.
- **Label:** the outcome or target we want to predict, if one exists.
- **Model:** a mathematical rule that turns input information into an output.
- **Training:** adjusting the model using past labeled examples.
- **Prediction:** using the trained model on a new example.
- **Evaluation:** checking whether the model performs well enough on data that were not used to fit it.

Not all data science is prediction. Some tasks are about grouping similar items, detecting anomalies, ranking search results, or summarizing patterns. Retrieval will later fall into the ranking family.

Formal version

A labeled dataset is often written as pairs (x_i, y_i) .

- x_i means the input for example i .
- y_i means the desired output for example i .

For spam filtering, x_i could be one email represented by features, and y_i could be the label *spam* or *not spam*.

Rule-based systems versus learned systems

A **rule-based system** uses instructions written directly by a human. A **learned system** uses data to estimate the instructions. Rule-based systems are often transparent and fast to build when the pattern is simple. Learned systems are often stronger when the pattern is messy, subtle, or too large to encode by hand.

Common mistake

A learned system is not magical. It still depends on human choices: what data were collected, which features were built, what metric was optimized, and what counts as

good performance.

Do not memorize this yet

Do not memorize the phrase “model” as something mysterious. For now, a model is just a function that takes input and produces an output. The sophistication comes later.

Self-check

1. In the spam example, name one possible feature and one possible label.
2. What is the difference between training and prediction?
3. Give one situation where a rule-based system might be enough.

Short answers

1. A feature could be the number of links; a label could be spam or not spam.
2. Training uses past examples to fit the model; prediction uses the fitted model on a new example.
3. A rule-based system might be enough when the pattern is simple and explicit, such as “reject files larger than a fixed size” or “flag messages containing an exact banned phrase.”

If you feel lost, remember this

Data science is the disciplined use of examples to build, compare, and evaluate useful decision rules. Retrieval will later become one special case: instead of predicting one label, we will score and rank many documents.

Text as data

Why this chapter exists

Retrieval works on text. But computers do not directly manipulate meaning in the way humans do. They manipulate symbols, indices, and numbers. This chapter explains how raw text becomes something a machine can compute with.

Intuition in ordinary language

When you read the sentence “the application crashes after startup,” you understand words, grammar, and likely context. A computer, at the beginning, only receives characters. To compare texts, count words, or compute similarity, the text must be transformed into a more structured representation.

Practical intuition

Think of preprocessing as preparing ingredients before cooking. The vegetables are still the vegetables, but they are washed, cut, and arranged so the next steps become possible and consistent.

Concrete toy example

Consider the raw text:

Wi-Fi/driver-error after Update!!!

A typical light normalization pipeline might do the following:

1. replace separators such as `or` or `-` with spaces,
2. lowercase everything,
3. remove repeated punctuation or ignore it,
4. split the text into tokens.

The result might be the token list:

```
[wi, fi, driver, error, after, update]
```

Technical explanation

Several beginner words must be separated carefully.

- **Token:** one unit the system decides to work with, often a word or word-piece.
- **Tokenization:** the process of splitting text into tokens.
- **Normalization:** making text more consistent, for example lowercasing or standardizing separators.
- **Preprocessing:** the larger preparation stage that may include normalization, tokenization, stopword handling, and other cleaning choices.
- **Stopwords:** very common function words such as *the*, *is*, and *and*. Depending on the method, they may be kept or removed.

Why do these choices matter? Because the machine only sees the representation we give it. If **Error-42**, **error 42**, and **ERROR/42** become three unrelated forms, the system may miss matches that a human would treat as obvious.

Tiny worked example

Suppose one query contains **startup freeze** and one document contains **Start-Up freezes**. Without some normalization, a lexical method may treat these as more different than necessary. With consistent lowercasing and separator handling, the overlap becomes easier to recover.

Formal view

If a preprocessing pipeline is written as $P(\cdot)$, then instead of using raw text t , the system uses $P(t)$. The important point is not the symbol itself. The important point is that the rest of the pipeline depends on the transformed version, not on the original raw string.

Common mistake

Preprocessing is not automatically “good” when it is aggressive. Too much cleaning can remove useful signal. For technical retrieval, numbers, symbols, version names, and file names can matter. A cleaning step that throws them away may damage performance.

Do not memorize this yet

Do not memorize a universal preprocessing recipe. There is no single best one. The right choice depends on the task, the data, and the retrieval model.

Self-check

1. What is the difference between a token and tokenization?
2. Why might lowercasing help a lexical retrieval system?
3. Give one case where removing too much text information could be harmful.

Short answers

1. A token is one unit such as a word or word-piece; tokenization is the act of splitting text into those units.
2. Lowercasing can make forms such as `Kernel` and `kernel` match more consistently.
3. Removing numbers or symbols could be harmful when error codes, version names, paths, or commands are important.

If you feel lost, remember this

Text retrieval starts with representation. The system cannot search meaning directly from raw strings; it first needs a workable form of the text.

Vectors, matrices, and similarity from zero

Why this chapter exists

Retrieval models compare texts. To compare them mathematically, they are often converted into vectors. This chapter gives the minimum linear-algebra intuition needed for the rest of the guide.

Intuition in ordinary language

A **vector** is an ordered list of numbers. Each position in the list stands for something. If the list describes a text, one position might stand for how much the word *kernel* matters, another for how much *update* matters, and so on. A **matrix** is a table of numbers built by stacking vectors.

Practical intuition

If a grocery list says apples = 2, milk = 1, bread = 0, then the list can be written as a vector. If many customers each have a grocery list written in the same order, all those lists together form a matrix.

Sparse versus dense

A **sparse vector** has mostly zeros. This often happens when each dimension corresponds to one vocabulary item and a document uses only a small fraction of the full vocabulary. A **dense vector** has many non-zero values, with meaning spread across the full pattern.

	Sparse representation	Dense representation
Typical dimensions	Often very large	Usually moderate and fixed
Most entries	Zero	Non-zero
Human interpretability	Often easier: one dimension may map to one token	Harder: one coordinate alone rarely has a simple meaning
Strength	Exact lexical signal	Semantic signal and compactness
Weakness	Poor on paraphrases	Harder to interpret; often more expensive to build

Dot product intuition

Suppose two vectors are

$$\mathbf{a} = (2, 1, 0), \quad \mathbf{b} = (3, 0, 4).$$

Their dot product is

$$\mathbf{a} \cdot \mathbf{b} = 2 \cdot 3 + 1 \cdot 0 + 0 \cdot 4 = 6.$$

The dot product becomes larger when the two vectors are strong in the same directions. In retrieval language, that often means the query and the document emphasize similar features.

Cosine similarity intuition

Raw size can be misleading. A very long document may have large counts simply because it has more words. Cosine similarity compares direction instead of raw magnitude:

$$\cos(\theta) = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\| \|\mathbf{d}\|}.$$

You do not need to love the formula yet. Read it this way: the numerator rewards shared signal, and the denominator rescales by vector lengths so we compare alignment rather than sheer size.

Tiny worked example

Let $\mathbf{q} = (1, 1)$, $\mathbf{d}_1 = (2, 2)$, and $\mathbf{d}_2 = (2, 0)$. Query and document d_1 point in exactly the same direction, so cosine similarity is 1. Document d_2 shares only part of the query direction, so its cosine similarity is smaller.

Formal version

A document-term matrix can be written as $X \in \mathbb{R}^{N \times V}$.

- N is the number of documents.
- V is the number of features or vocabulary dimensions.

- Row i stores the vector representation of document i .

In sparse lexical models, V can be very large. In dense embedding models, V is replaced by a smaller fixed dimension such as 384.

Common mistake

Students often confuse “vector” with “semantic embedding.” An embedding is one kind of vector, not the definition of vector itself. TF-IDF vectors are vectors too.

Self-check

1. Why is a bag-of-words representation usually sparse?
2. What problem does cosine similarity try to reduce?
3. If two vectors point in almost the same direction, what should happen to cosine similarity?

Short answers

1. Because the vocabulary is large but each text uses only a small fraction of all possible terms, so most entries are zero.
2. It reduces the effect of raw vector length so long texts do not automatically dominate.
3. Cosine similarity should be high, because the vectors are well aligned.

If you feel lost, remember this

Similarity is central because retrieval is not just “does this document contain a word?” It is “how strongly does this document align with the query representation?”

Part II

Understanding the retrieval problem

What information retrieval is

Why this chapter exists

Now that you know how data and text can be represented, you are ready to understand the retrieval problem itself: given a query, how do we search a large corpus and order the documents so the most helpful ones appear first?

Intuition in ordinary language

Information retrieval is the science of finding the most relevant items from a large collection. Search engines, digital libraries, legal search tools, product search, code search, and technical forum search are all retrieval systems.

A retrieval system does not usually return just one answer. It returns a **ranking**: a list ordered from most promising to least promising.

Practical intuition

If you type “android app crashes on startup” into a technical forum search system, the system may search hundreds of thousands of posts. The challenge is not only finding matching posts. It is ordering them so the useful ones appear early.

Core objects

- **Corpus**: the full collection of searchable documents.
- **Document**: one item in the corpus.
- **Query**: the user-side request or problem statement.
- **Relevance**: whether a document actually helps answer the query.
- **Ranking**: the ordered list produced by the system.
- **Top- k** : the first k documents kept or evaluated.

Formal version

If the corpus is $D = \{d_1, d_2, \dots, d_N\}$ and the query is q , a retriever computes a score for each document and sorts them:

$$\text{score}(q, d) \in \mathbb{R}, \quad d \in D.$$

The final ranking is the documents arranged from highest score to lowest score.

Retrieval is not classification

These two tasks are often confused.

Aspect	Retrieval	Classification
Input	One query plus an entire corpus	One example at a time
Output	Ranked list of documents	One class label or class probabilities
Main question	Which items are most relevant?	Which category does this item belong to?
Typical metric	Recall, precision, MRR	Accuracy, F1, calibration

Concrete mini example

Suppose the query is “latex bibliography stopped compiling.”

- Retrieval asks: which posts in the forum should be shown first?
- Classification asks: is this query about *tex*, *programmers*, *android*, *gaming*, or *unix*?

These are different jobs. Later, classification will help retrieval, but it will not replace it.

Common mistake

A relevant document does not need to share many exact words with the query. A document can be relevant because it describes the same issue using different language.

Self-check

1. What is the difference between a corpus and one document?
2. What does top- k mean?
3. Why is retrieval usually about ranking rather than a single yes/no decision?

Short answers

1. The corpus is the whole searchable collection; a document is one item inside it.
2. Top- k means the first k highest-ranked results.

3. Because the system often needs to order many candidate documents by usefulness, not just decide whether one document is acceptable.

If you feel lost, remember this

Retrieval is the task of ranking a corpus against a query. The central problem is not classification, but ordered relevance.

Lexical matching versus semantic matching

Why this chapter exists

The biggest conceptual divide in modern retrieval is the divide between *lexical* matching and *semantic* matching. Understanding that divide will make the later chapters on TF-IDF, BM25, embeddings, and rerankers much easier.

Intuition in ordinary language

Lexical matching cares about words or tokens that literally overlap. Semantic matching cares about closeness in meaning, even when the exact words differ.

Tiny worked example

Query: *program hangs during launch.*

Document A: *program hangs during launch.*

Document B: *application freezes on startup.*

A lexical system strongly likes Document A because the wording overlaps exactly. A semantic system may still like Document B because *program* and *application*, or *launch* and *startup*, are similar in meaning.

Vocabulary mismatch

Vocabulary mismatch means the query and the relevant document talk about the same thing but use different words. This is one of the main reasons semantic retrieval became important.

Practical intuition

If one user searches for *laptop battery drains quickly* and the document says *notebook loses charge fast*, the meaning is close but the lexical overlap is weak.

When lexical methods win

Lexical methods are strong when exact tokens matter, for example:

- error codes,
- command names,
- API names,
- package names,
- configuration flags,
- exact technical terms.

If a query contains a rare exact string, lexical retrieval often has a major advantage.

When semantic methods win

Semantic methods are strong when:

- the query is a paraphrase,
- synonyms are common,
- the wording differs across authors,
- the user describes the problem rather than quoting exact terms.

Formal version

You can think of the contrast this way:

- Lexical retrieval estimates relevance from token overlap and token weighting.
- Semantic retrieval estimates relevance from closeness in a learned meaning space.

The formal machinery differs, but the practical distinction is already enough to understand why the two families behave differently.

Common mistake

Do not turn lexical versus semantic into a fight with one winner. Real retrieval systems often combine them because they fail in different ways.

Self-check

For each situation, say whether lexical matching, semantic matching, or both are likely to help most.

1. The query contains a rare error code.
2. The query is a paraphrase of the document title.

3. A user searches for a package name and a plain-language description together.

Short answers

1. Lexical matching is usually very strong because exact overlap matters.
2. Semantic matching often helps because the wording differs.
3. Both may help: lexical for the exact package name and semantic for the descriptive part.

If you feel lost, remember this

Lexical methods ask “which words overlap?” Semantic methods ask “which meanings are close?” Modern retrieval uses both because users express the same problem in many different ways.

Part III

Lexical retrieval models

Bag of words from zero

Why this chapter exists

Most classical lexical retrieval methods begin with the bag-of-words view of text. It is simple, imperfect, and foundational.

Intuition in ordinary language

Bag of words treats a text as a multiset of tokens. The order of words is mostly ignored. What matters is which tokens appear and how often they appear.

Tiny worked example

The sentence *kernel panic after update* becomes counts such as:

kernel : 1, panic : 1, after : 1, update : 1.

The sentence *update after kernel panic* would have the same bag-of-words counts, even though the original order changed.

Vocabulary and document-term view

Choose a vocabulary, for example:

$$V = [\text{kernel}, \text{panic}, \text{update}, \text{driver}].$$

Then each document becomes a vector of counts. If a document contains *kernel panic after update*, its vector might be $(1, 1, 1, 0)$.

If you stack many such vectors, you get a document-term matrix.

Why this view is useful

Bag of words is useful because it is easy to build, easy to inspect, and often very strong on technical corpora where exact wording matters.

Why it is limited

Because order is mostly ignored, bag of words cannot fully represent phrases, syntax, negation, or broader meaning. *Install driver* and *Do not install driver* can look too similar in a crude count-based representation.

Common mistake

Bag of words is not “wrong.” It is a deliberate simplification. Many strong systems start from it because a useful simplification can outperform a complicated model when the task favors exact terms.

Self-check

1. Why is the model called “bag” of words?
2. What important information is largely lost in this representation?
3. Why can it still work well for technical retrieval?

Short answers

1. Because it keeps the collection of words and their counts while largely ignoring order.
2. Word order, syntax, and finer meaning are largely lost.
3. Because exact technical tokens often carry strong signal, and count-based overlap can already be very informative.

If you feel lost, remember this

Bag of words is the simplest way to turn text into count-based vectors. Later methods such as TF-IDF and BM25 improve how those counts are weighted.

TF, DF, and IDF slowly and properly

Why this chapter exists

Raw counts are a start, but they are not enough. This chapter explains the three pieces that lead into TF-IDF: term frequency, document frequency, and inverse document frequency.

Intuition in ordinary language

If a term appears many times in a document, it probably matters for that document. But if the same term appears in almost every document, it is not very discriminative. TF captures local importance inside one document. IDF captures how informative the term is across the whole corpus.

Concrete toy corpus

Suppose the corpus contains three documents:

- d_1 : *kernel panic after update*
- d_2 : *wifi driver update issue*
- d_3 : *kernel module compilation*

Now compare the terms *kernel* and *panic*.

- *kernel* appears in d_1 and d_3 , so its document frequency is 2.
- *panic* appears only in d_1 , so its document frequency is 1.

This means *panic* is rarer and thus more discriminative.

Technical explanation

- **Term frequency (TF)**: how often a term appears inside one document.
- **Document frequency (DF)**: in how many documents the term appears at least once.
- **Inverse document frequency (IDF)**: a weight that grows when the term is rare across the corpus.

Formal version

If the corpus has N documents and term t appears in $df(t)$ of them, a common IDF form is

$$\text{idf}(t) = \log\left(\frac{N}{df(t)}\right).$$

Read the formula in plain language:

- If $df(t)$ becomes larger, the denominator grows, so IDF becomes smaller.
- If $df(t)$ becomes smaller, the term is rarer, so IDF becomes larger.

The exact formula may be smoothed differently in different libraries, but the intuition stays the same.

What problem does this solve?

If you only use raw counts, very common words can dominate the representation. IDF reduces the influence of terms that appear almost everywhere and gives more value to terms that actually distinguish one document from another.

Common mistake

Students sometimes say “TF-IDF is just repeated-word counting.” That is incomplete. Repetition matters, but rarity across the corpus matters too.

Self-check

1. Which term should get a higher IDF: a term appearing in 2 documents out of 1,000 or a term appearing in 900 documents out of 1,000?
2. Why does document frequency ignore repeated appearances inside the same document?
3. In the toy corpus above, which term is more discriminative: *kernel* or *panic*?

Short answers

1. The term appearing in only 2 documents should get the higher IDF, because it is rarer.
2. Because DF asks in how many documents the term occurs at least once, not how many times overall.
3. *panic* is more discriminative because it appears in fewer documents.

If you feel lost, remember this

TF tells you how prominent a term is inside one document. IDF tells you how informative that term is across the corpus. Retrieval needs both ideas.

TF–IDF for retrieval

Why this chapter exists

This chapter brings the previous pieces together. TF–IDF is the classical sparse lexical weighting scheme that many students meet first in information retrieval and text mining.

From zero: what problem does TF–IDF solve?

Raw counts do not distinguish common terms from informative ones. TF–IDF says: reward a term when it is frequent in this document but not frequent everywhere.

Formal version

A simple TF–IDF weight for term t in document d is

$$\text{tfidf}(t, d) = \text{tf}(t, d) \cdot \text{idf}(t).$$

Read this slowly:

- If the term appears more often in document d , the TF part grows.
- If the term appears in many documents, the IDF part shrinks.
- The product becomes large when the term is both locally important and globally informative.

Query and document vectors

The query also becomes a TF–IDF vector in the same feature space as the documents. Then we compare the query vector with each document vector, commonly using cosine similarity.

Tiny worked example

Suppose the query is *kernel panic*. A document containing both *kernel* and *panic*, especially if *panic* is rare in the corpus, will often receive a high TF–IDF cosine score.

Strengths

- fast and simple,
- easy to inspect,
- strong on exact technical terms,
- strong as a baseline and debugging tool.

Weaknesses

- weak on paraphrases,
- weak when synonyms replace the query terms,
- still driven by token overlap,
- can under-represent deeper meaning.

Small scoring intuition

If query and document share a rare informative term, the cosine score rises strongly. If they share only very common terms, the score rises much less. If there is no direct overlap, lexical TF-IDF gives little or no signal.

Common mistake

A low TF-IDF score does not prove a document is irrelevant. It may simply mean the wording differs. That is one of the main motivations for dense semantic retrieval.

Self-check

1. Why is TF-IDF usually called a sparse representation?
2. Why does cosine similarity often pair naturally with TF-IDF?
3. Which failure mode is most natural for TF-IDF: exact rare-code mismatch or paraphrase mismatch?

Short answers

1. Because the vector has one dimension per vocabulary item and most entries are zero.
2. Because cosine similarity compares the alignment of weighted vectors rather than raw length.
3. Paraphrase mismatch is the more natural failure mode, because the wording can differ even when the meaning matches.

If you feel lost, remember this

TF-IDF is a weighted lexical overlap model: it does more than count words, but it still depends mainly on direct term overlap.

BM25

Why this chapter exists

BM25 exists because plain TF-IDF does not explicitly model two practical realities well enough: repeated term occurrences should help with diminishing returns, and document length should be corrected more directly.

Intuition in ordinary language

If a relevant term appears once in a document, that is useful evidence. If it appears five times, that is stronger evidence. But the fifth occurrence is not five times as informative as the first. Also, long documents should not win automatically just because they contain more words overall.

Practical intuition

Imagine two documents that both mention *kernel panic*. One is a short focused troubleshooting post. The other is a giant mixed discussion thread that mentions *kernel* many times among many unrelated details. BM25 tries to reward the first kind of document more sensibly.

Term-frequency saturation

BM25 does not let evidence grow linearly forever. The first few occurrences help a lot; later repetitions help less and less. This idea is called **term-frequency saturation**.

Length normalization

BM25 also adjusts the term-frequency contribution according to document length. A very long document is not allowed to dominate simply because it has more opportunities to mention a term.

Formal version

A common BM25 score is

$$\text{score}(q, d) = \sum_{t \in q} \text{idf}(t) \frac{f(t, d)(k_1 + 1)}{f(t, d) + k_1 \left(1 - b + b \frac{|d|}{\text{avgdl}}\right)}.$$

Read the symbols slowly.

- $f(t, d)$ is how many times term t occurs in document d .
- $|d|$ is the length of document d .
- avgdl is the average document length in the corpus.
- k_1 controls how quickly the term-frequency reward saturates.
- b controls how strongly length normalization matters.

If $f(t, d)$ gets larger, the score increases, but with diminishing returns. If $|d|$ becomes much larger than average and the other evidence stays the same, the denominator grows and the score is moderated.

Why BM25 can feel more natural than TF-IDF

BM25 was built for retrieval. It handles lexical overlap with a scoring rule that reflects common retrieval intuitions more directly than raw TF-IDF cosine similarity.

Common mistake

BM25 is not a tokenizer and not a stopword remover. Tokenization and stopword handling can happen before BM25, but BM25 itself is a scoring formula.

Self-check

1. What does term-frequency saturation mean in plain language?
2. Why does BM25 care about document length?
3. If two documents contain the same important term equally often, which one is BM25 more cautious about: a short document or a much longer document?

Short answers

1. It means repeated occurrences help, but each additional repetition helps less than the previous one.
2. Because long documents would otherwise get unfair advantage simply for having more words.
3. BM25 is more cautious about the much longer document.

If you feel lost, remember this

BM25 is still lexical, but it models repetition and length more realistically than plain TF-IDF.

BM25+

Why this chapter exists

BM25+ is a refinement of BM25. It keeps the main logic of BM25 but adds a small offset so that long documents are not overly penalized when they do contain the query terms.

Intuition in ordinary language

BM25 already improves lexical scoring, but it can still punish some long documents too harshly. BM25+ adds a small constant term, often called δ , to soften that problem.

Formal version

A common BM25+ form is

$$\text{score}(q, d) = \sum_{t \in q} \text{idf}(t) \left(\frac{f(t, d)(k_1 + 1)}{f(t, d) + k_1 \left(1 - b + b \frac{|d|}{\text{avgdl}} \right)} + \delta \right).$$

The new symbol is δ .

- If δ increases, matching terms get a slightly larger baseline contribution.
- The purpose is not to change BM25 into a new family. The purpose is to reduce over-penalization in some settings, especially with long documents.

Side-by-side comparison

Criterion	TF-IDF	BM25	BM25+
Main family	Sparse lexical	Sparse lexical	Sparse lexical
Exact overlap required	Yes	Yes	Yes
Term repetition handling	Present but indirect	Explicit saturation	Explicit saturation plus offset
Length correction	Mostly via cosine normalization	Explicit in the score	Explicit in the score with extra offset
Typical role	Strong baseline	Stronger lexical baseline	Slight refinement of BM25
Paraphrase handling	Weak	Weak	Weak

When BM25+ may help

If the corpus contains long documents whose relevant terms should still earn a solid base contribution, BM25+ can sometimes behave a little better than BM25. The gain is usually evolutionary, not revolutionary.

Common mistake

Do not explain BM25+ as “BM25 with better cleaning.” The important difference is in the lexical scoring formula, not in token cleanup.

Self-check

1. What new ingredient does BM25+ add compared with BM25?
2. Does BM25+ solve paraphrase mismatch?
3. Are TF-IDF, BM25, and BM25+ all sparse lexical methods or not?

Short answers

1. It adds a positive offset δ inside the term contribution.
2. No. It is still lexical and still depends mainly on overlap.
3. Yes. All three are sparse lexical retrieval methods.

If you feel lost, remember this

BM25+ is best understood as a careful refinement of BM25, not as a completely different family of retrieval models.

Part IV

Semantic retrieval models

Embeddings from zero

Why this chapter exists

Sparse lexical methods are powerful, but they struggle when relevant documents use different wording. Embeddings were developed to represent text in a dense meaning-oriented space where similar texts can land near one another even without strong token overlap.

Intuition in ordinary language

An embedding is a dense vector that tries to place semantically similar texts close together. Instead of giving one coordinate to one token, the model learns a compact pattern of numbers whose full shape carries meaning.

Tiny worked example

Suppose one query says *program hangs during launch* and a document says *application freezes on startup*. A lexical model sees weak direct overlap. A good embedding model can still map the two texts to nearby vectors because their meanings are related.

Why dense vectors can represent meaning

The coordinates of an embedding are learned from large amounts of training data. The model adjusts its internal parameters so that texts appearing in similar contexts or with similar supervision end up near each other. One coordinate by itself usually has no simple human interpretation. The semantic signal lives in the entire pattern.

Formal version

If $e(x)$ denotes the embedding of text x , then a document and a query become vectors such as

$$e(q) \in \mathbb{R}^d, \quad e(d) \in \mathbb{R}^d.$$

Here d is a fixed dimension, for example 384. The exact number matters less than the idea that every text is mapped into the same dense vector space.

Why embeddings are different from token counts

TF-IDF might say “this dimension means the token *kernel*.” Embeddings do not work that way. Their coordinates are learned jointly. That makes them powerful for meaning, but less transparent.

Common mistake

Do not say “embeddings are just another way to count words.” They are learned dense representations, not token-count tables.

Self-check

1. Why are embeddings called dense?
2. Why might one coordinate of an embedding be hard to interpret alone?
3. What kind of mismatch are embeddings especially good at reducing?

Short answers

1. Because many entries are non-zero and meaning is distributed across the whole vector.
2. Because the semantic information is encoded in the full learned pattern, not in one coordinate with an obvious label.
3. They are especially good at reducing vocabulary or paraphrase mismatch.

If you feel lost, remember this

An embedding is a dense learned vector whose main job is to place semantically related texts near each other.

Bi-encoder retrieval

Why this chapter exists

Bi-encoder retrieval is the main large-scale semantic retrieval pattern. It explains why dense retrieval can be used over a big corpus without rereading every query-document pair jointly.

Intuition in ordinary language

A bi-encoder has two encoding paths: one for the query and one for the document. Each side is encoded separately. After that, similarity can be computed between the resulting vectors.

Practical intuition

The great efficiency trick is this: documents can be encoded once and stored. When a new query arrives, only the query needs to be encoded on the fly, and then it can be compared with the stored document embeddings.

Technical explanation

Let $e_q(q)$ be the query encoder and $e_d(d)$ be the document encoder. Many practical systems use the same underlying model family for both. The retrieval score may be cosine similarity or dot product between the two vectors.

Formal version

A bi-encoder score can be written as

$$\text{score}(q, d) = e_q(q)^\top e_d(d)$$

or, after normalization, as cosine similarity. The important point is that q and d are encoded *separately*.

Why this is efficient

Because the expensive document encoding step is moved offline. Large corpora become searchable through vector similarity search rather than full pairwise neural scoring at query time.

Strengths and weaknesses

- strong on paraphrase and semantic similarity,
- scalable because document vectors can be reused,
- weaker than a cross-encoder at very fine pairwise judgment,
- can retrieve semantically plausible but not truly relevant items.

Common mistake

Bi-encoder retrieval is not the same as “semantic retrieval in general.” It is one scalable architecture for semantic retrieval.

Self-check

1. Why can document embeddings be cached in a bi-encoder system?
2. Which is usually more scalable over a large corpus: bi-encoder retrieval or cross-encoder scoring of every pair?
3. What is the main trade-off of the bi-encoder design?

Short answers

1. Because documents are encoded independently of the query, so the same document embedding can be reused for many future queries.
2. Bi-encoder retrieval is much more scalable.
3. It gains speed and scalability, but usually loses some pairwise precision compared with a cross-encoder.

If you feel lost, remember this

Bi-encoders make semantic retrieval practical at scale by separating query encoding from document encoding.

Limits of embeddings

Why this chapter exists

Dense retrieval is powerful, but it is not free and it is not perfect. Understanding its limitations prevents unrealistic expectations and explains why later reranking or category-aware adjustment may help.

Computational cost

Encoding a large corpus into dense vectors can be expensive. Scoring a query against all document vectors also consumes memory and compute. This is why caching, batching, approximate search, and chunking matter in real systems.

Memory cost

A large dense matrix can occupy substantial memory. Even when each vector dimension is modest, the corpus may contain hundreds of thousands or millions of documents.

Semantically plausible but wrong

Embeddings can retrieve documents that sound related but miss the exact issue. For example, a query about *Android package installation* may retrieve a document about *mobile app deployment* that is semantically close but belongs to the wrong subtopic.

Category confusion

Semantic closeness does not guarantee the right topical region. This is one reason a category classifier or a reranker can still help.

Practical intuition

Dense retrieval often behaves like a generous first reader: it brings many meaning-related candidates to the table. A later stage may still be needed to decide which of those candidates are most precisely relevant.

Recall versus precision

Dense retrieval often improves **recall**: more relevant items enter the candidate set. But some extra noise can also enter, so first-stage precision may not improve by the same amount.

Common mistake

Do not think “dense retrieval understands meaning perfectly.” It learns useful semantic geometry, but it can still confuse neighboring topics, over-generalize, or miss exact lexical constraints.

Self-check

1. Why can dense retrieval increase recall?
2. Why might it also add noise?
3. Name one engineering consequence of dense retrieval.

Short answers

1. Because it can recover semantically similar documents even when exact wording differs.
2. Because semantic closeness does not always mean exact relevance, so plausible but wrong items can enter the ranking.
3. It creates heavier compute and memory requirements, so caching and efficient scoring become important.

If you feel lost, remember this

Dense retrieval is often an excellent candidate generator, but it is not usually the final word on precision.

Part V

Reranking and deeper relevance

What a reranker is

Why this chapter exists

First-stage retrieval must be fast. Because of that speed requirement, it usually makes some mistakes near the top of the ranking. A reranker exists to make better local decisions on a smaller candidate set.

Intuition in ordinary language

A reranker is a second-stage model. It does not search the full corpus from scratch. Instead, it takes the candidate documents already retrieved by the first stage and reorders them more carefully.

Tiny worked example

Suppose the first-stage retriever returns 1,000 candidates for a query. A reranker might inspect only the best 50 or 100 of those and decide that the second item should really be first, the fifth should move to third, and so on.

Candidate set intuition

The candidate set is the short list that survives the fast retrieval stage. The main job of the first stage is good coverage. The main job of the reranker is good local ordering.

Formal version

If the first-stage retriever returns the top- k candidates

$$C_k(q) = \{d_{i_1}, d_{i_2}, \dots, d_{i_k}\},$$

a reranker computes a new score only on that set, not on the entire corpus.

Why reranking is local rather than full-corpus search

A sophisticated neural model may be too slow to run on every query-document pair in a huge corpus. Running it only on a shortlist gives a good trade-off: the fast retriever narrows the space, and the reranker spends compute where it matters most.

Common mistake

Reranking is not “another retrieval model doing the same job.” It is a second-stage refinement step operating on candidates that have already been found.

Self-check

1. Why is a reranker usually applied after retrieval rather than before it?
2. What is the candidate set?
3. Which stage usually cares more about speed over the whole corpus: first-stage retrieval or reranking?

Short answers

1. Because reranking models are often too expensive to score the whole corpus directly.
2. It is the shortlist of retrieved documents passed to the second stage.
3. First-stage retrieval usually has the stricter whole-corpus speed requirement.

If you feel lost, remember this

The first stage finds plausible candidates; the reranker tries to decide which of those plausible candidates deserve the top positions.

Cross-encoder from zero

Why this chapter exists

The cross-encoder is the most important reranking concept in modern retrieval pipelines. It explains how a model can read a query and a document together and make a more precise pairwise relevance judgment than a bi-encoder usually can.

Intuition in ordinary language

A bi-encoder reads query and document separately. A cross-encoder reads the *pair jointly*. Because it sees both texts together, it can model detailed interactions between them.

Tiny worked example

Query: *latex references stopped compiling.*

Candidate document 1: *bibtex error after package update.*

Candidate document 2: *python package install failed.*

A cross-encoder can use the whole pair context to give the first candidate a higher pairwise relevance score, even if both candidates contained the word *package*.

Technical explanation

The cross-encoder produces one score for one query-document pair. It is usually applied only to a small reranking depth because it must process each pair separately.

Formal version

A cross-encoder score can be written abstractly as

$$\text{score}(q, d) = f_{\theta}(q, d),$$

where the model sees both texts jointly. The important difference from the bi-encoder is that there are no separately cached document vectors that can be reused for arbitrary

similarity comparison in the same way.

Bi-encoder versus cross-encoder

Aspect	Bi-encoder	Cross-encoder
How texts are processed	Separately	Jointly as a pair
Scalability over a large corpus	High	Low
Can document vectors be cached?	Yes	Not in the same direct similarity-search way
Typical role	First-stage retrieval	Reranking
Pairwise precision	Good	Often better

Before-and-after ranking intuition

If the bi-encoder retrieved candidates in the order d_2, d_1, d_3 , a cross-encoder might rescore them and reorder them as d_1, d_2, d_3 . That local correction can improve metrics such as reciprocal rank and MRR.

Common mistake

Students sometimes ask why we do not run the cross-encoder on the whole corpus. The answer is computational: doing joint neural scoring for every query-document pair in a large corpus is usually far too expensive.

Self-check

1. What does it mean that a cross-encoder reads a query and a document jointly?
2. Why is it usually used only on a shortlist?
3. Which architecture is usually better for pairwise precision: bi-encoder or cross-encoder?

Short answers

1. It means the model processes the two texts together as one pair and can reason about their interactions directly.
2. Because pairwise joint scoring is expensive over a large corpus.
3. The cross-encoder is usually better for pairwise precision.

If you feel lost, remember this

Bi-encoders are efficient candidate generators. Cross-encoders are slower but often sharper judges of which candidate is truly most relevant.

Hard negatives and training intuition

Why this chapter exists

To train a reranker well, it is not enough to show it only obviously wrong examples. It must also learn from difficult mistakes. That is where hard negatives come in.

Positive and negative pairs

A positive pair is a query and a relevant document. A negative pair is a query and a non-relevant document. The reranker learns to score positives above negatives.

Tiny worked example

Positive pair: query *android app crashes on startup* with a document about that same crash issue.

Easy negative: the same query paired with a document about *latex bibliography formatting*.

Hard negative: the same query paired with a document about *android app freezes while loading images*. The topic is close, but the exact issue may still be wrong.

Why hard negatives matter

If all negatives are easy, the model learns only to reject absurd mismatches. It does not become very good at the harder and more important job: separating a truly relevant candidate from a plausible near-miss.

Practical intuition

A student preparing for an exam improves more by practicing confusing near-cases than by answering questions whose wrong options are ridiculous. Hard negatives are the retrieval equivalent of those near-cases.

Training intuition

During training, the reranker is encouraged to give higher scores to positive pairs and lower scores to negative pairs. The exact loss function can vary, but the practical idea is stable: reward correct relative ordering, punish incorrect relative ordering.

Formal version

One simple binary classification view is

$$L(q, d, y) = -y \log \hat{y} - (1 - y) \log(1 - \hat{y}),$$

where $y \in \{0, 1\}$ is the true label and $\hat{y}$ is the predicted relevance probability or score after transformation. You do not need to memorize the loss. The main idea is that wrong predictions are penalized.

Common mistake

Hard negatives are not random wrong documents from the whole corpus. They are usually mined from documents that already looked plausible to the first-stage retriever.

Self-check

1. What is the difference between an easy negative and a hard negative?
2. Why are hard negatives especially valuable for a reranker?
3. In one sentence, what is the training objective trying to achieve?

Short answers

1. An easy negative is obviously unrelated; a hard negative is plausible and close to the query but still not truly relevant.
2. Because reranking needs to distinguish relevant documents from confusing near-misses, not only from absurd mismatches.
3. The objective is trying to make relevant pairs score higher than non-relevant pairs.

If you feel lost, remember this

A reranker becomes useful not merely by learning what is wrong in general, but by learning what is almost right and still wrong.

Part VI

Classification inside a retrieval pipeline

What classification is

Why this chapter exists

Classification appears inside the retrieval pipeline later as a helper. Before using it that way, it must be understood on its own terms.

Intuition in ordinary language

Classification assigns an item to one category from a fixed set of categories. If the categories are *android*, *unix*, *tex*, *gaming*, and *programmers*, a classifier tries to predict which one best fits a given query or document.

Input and output

Classification takes one text at a time and outputs a class label or scores over labels. Retrieval takes one query and a large corpus and outputs a ranking of documents. That difference is fundamental.

Tiny worked example

Input text: *How do I fix a BibTeX compilation problem after updating TeX Live?*
Likely classification output: *tex*.

Why classification is useful in a retrieval project

Categories can act like topical regions. If a query is very likely *android*, then documents from the *android* category may deserve slightly more attention during reranking.

Formal version

In single-label classification, a model maps one input x to one predicted label $\hat{c}$ from a fixed set of classes $\mathcal{C}$:

$$\hat{c} = g(x), \quad \hat{c} \in \mathcal{C}.$$

Common mistake

Do not say “classification finds relevant documents.” Classification predicts categories. Relevance is a richer notion than category membership.

Self-check

1. What is the output of a classifier?
2. Why is category not the same thing as relevance?
3. Why can a classifier still be useful inside a retrieval pipeline?

Short answers

1. A class label or class scores for one input example.
2. Because many documents can share a category but still differ strongly in relevance to a specific query.
3. Because category can provide a useful topical prior or soft signal for reranking.

If you feel lost, remember this

Classification and retrieval are different tasks. In this guide, classification is a helper that can improve ranking slightly, not the main retrieval mechanism.

Why classification can help retrieval

Why this chapter exists

This chapter explains the exact role of classification inside a retrieval pipeline: not replacing retrieval, but gently guiding it.

Category prior intuition

If a query is predicted to belong to *android*, then an *android* document is often more plausible than a *tex* document, all else equal. This is a prior: a weak expectation about where relevance is more likely to lie.

Soft bonus versus hard filtering

There are two common ways to use that prior.

Method	What it does	Main risk
Hard filtering	Discards documents whose category does not match the predicted query category	If the classifier is wrong, relevant documents may disappear completely
Soft bonus	Keeps all candidates but adds a small score bonus to category matches	Can still bias the ranking, but less destructively

Why the soft bonus is safer

A soft bonus says: “category agreement is a helpful hint, not a law.” The main relevance signal still comes from the retrieval score or reranker score.

Tiny worked example

Suppose the predicted category is *android*. If two documents have similar retrieval scores, the *android* one may receive a small bonus and move higher. But a very relevant *programmers* document is not automatically deleted.

Formal view

A simple score adjustment is

$$\text{final_score}(q, d) = \tilde{s}(q, d) + \lambda \mathbf{1}[\hat{c}(q) = c(d)],$$

where $\tilde{s}(q, d)$ is a normalized base score, λ is the bonus strength, and the indicator is 1 when the predicted query category matches the document category and 0 otherwise.

If λ becomes larger, category agreement matters more. If $\lambda = 0$, the classifier has no effect on ranking.

Common mistake

A classifier is only a helper. If the category bonus becomes too strong, it can distort relevance instead of improving it.

Self-check

1. Why is hard filtering riskier than a soft bonus?
2. What does the parameter λ control?
3. Why might a category-matching document still be irrelevant?

Short answers

1. Because a wrong category prediction can remove relevant documents entirely.
2. It controls how much score is added when the predicted category matches the document category.
3. Because category is only a broad topical region, not a guarantee of exact relevance.

If you feel lost, remember this

Category is a weak prior, not the definition of relevance. That is why a soft bonus is usually safer than a hard filter.

TF-IDF + LinearSVC from zero

Why this chapter exists

A very common classical text-classification pipeline uses TF-IDF features with a linear classifier such as LinearSVC. This chapter explains what that means in beginner language.

Intuition in ordinary language

The input text is first converted into a TF-IDF vector. Then the classifier learns one linear decision rule per class. Each class gets a score, and the class with the highest score is predicted.

Tiny worked example

A query containing terms such as *latex*, *bibtex*, and *compile* may push the *tex* class score upward. A query containing *android*, *apk*, and *emulator* may push the *android* class score upward.

What “linear” means here

A linear model combines feature values with learned weights. If a feature is strongly associated with one class, its weight for that class becomes large and positive. If it works against that class, the weight can become negative.

Formal version

A linear class score for class c can be written as

$$\text{score}_c(x) = \mathbf{w}_c^\top \mathbf{x} + b_c,$$

where $\mathbf{x}$ is the TF-IDF vector of the text, $\mathbf{w}_c$ is the learned weight vector for class c , and b_c is a bias term. The predicted class is the one with the largest score.

Why this is called a classical text classifier

It does not rely on joint neural text understanding. It uses sparse engineered features plus a strong linear decision rule. That combination is often surprisingly effective, fast, and interpretable.

Why it fits retrieval pipelines well

Because TF-IDF features are already natural in text systems, and the classifier can often be trained and applied cheaply compared with heavier neural models.

Common mistake

Do not confuse LinearSVC with retrieval scoring. It predicts categories; it does not directly rank the corpus for a query.

Common mistake

High classification accuracy does not automatically mean the experimental design is clean. If extra fields such as tags, metadata tokens, or site-specific labels are almost unique to one class, the model may be reading the label indirectly. In that case the right response is to report that representation as label-revealing and keep it separate from the main pipeline.

Self-check

1. What does the input vector $\mathbf{x}$ represent in this pipeline?
2. What does it mean if a term has a strong positive weight for one class?
3. Why is this pipeline called classical rather than a neural semantic model?

Short answers

1. It is the TF-IDF feature vector of the text.
2. It means that term strongly supports that class when it appears.
3. Because it relies on sparse features and linear scoring rather than learned dense semantic representations of text pairs.

If you feel lost, remember this

TF-IDF + LinearSVC is a reliable, efficient way to predict text categories. In a retrieval pipeline, its output is best used as a supporting signal rather than the main ranking engine.

Part VII

Evaluation and experimental honesty

Evaluation metrics from zero

Why this chapter exists

A retrieval system cannot be improved honestly without good evaluation. This chapter explains the core metrics slowly, first in plain language and then with formulas.

Plain-language meaning of the main metrics

- **Precision:** among the documents we returned, how many were actually relevant?
- **Recall:** among all truly relevant documents, how many did we successfully return?
- **F1:** a balance between precision and recall.
- **Reciprocal rank:** how early did the first relevant document appear?
- **MRR:** average reciprocal rank over many queries.
- **Accuracy:** for classification, how often was the predicted class correct?

Tiny ranking example

Suppose a query has three relevant documents:

$$R_q = \{d_3, d_8, d_{20}\}.$$

Suppose the system returns this top-5 list:

$$[d_8, d_2, d_3, d_{11}, d_{40}].$$

The hits are d_8 and d_3 , so there are 2 hits in the top-5.

Formal version

If $\hat{R}_{q,K}$ is the set of returned top- K documents, then

$$\text{Precision@K} = \frac{|R_q \cap \hat{R}_{q,K}|}{K}, \quad \text{Recall@K} = \frac{|R_q \cap \hat{R}_{q,K}|}{|R_q|}.$$

In the example above,

$$\text{Precision@5} = \frac{2}{5} = 0.4, \quad \text{Recall@5} = \frac{2}{3} \approx 0.67.$$

If precision rises while recall stays fixed, the returned list became cleaner. If recall rises while precision stays fixed, the system recovered more relevant material without adding proportionally more noise.

F1

F1 is the harmonic mean of precision and recall:

$$F1 = \frac{2PR}{P + R}.$$

It is useful when you want one balanced number, but it hides the trade-off between precision and recall, so the components should still be inspected separately.

Reciprocal rank and MRR

If the first relevant document appears at rank r_q , then

$$\text{RR}(q) = \frac{1}{r_q}.$$

If the first relevant result is at rank 1, reciprocal rank is 1. If it appears at rank 5, reciprocal rank is 0.2. MRR is the average of that value across queries.

Accuracy

For classification, accuracy is simply the fraction of inputs whose predicted label matches the true label.

Metric	Main question	Sensitive to	Best when
Precision	How clean are the returned results?	False positives in the returned list	You care about avoiding noisy results
Recall	How much relevant material did we recover?	Missed relevant items	You care about coverage
MRR	How early is the first relevant result?	First-hit position	You care strongly about early ranking quality

Common mistake

Low precision does not always mean failure. If K is extremely large, precision can look numerically small even when the system has strong recall and decent early ranking.

Self-check

A query has relevant set $\{d_1, d_4\}$ and the top-4 returned documents are $[d_3, d_4, d_9, d_1]$.

1. Compute Precision@4.
2. Compute Recall@4.
3. What is the reciprocal rank?

Short answers

There are 2 hits in the top-4: d_4 and d_1 . So $\text{Precision@4} = 2/4 = 0.5$. $\text{Recall@4} = 2/2 = 1.0$. The first relevant result appears at rank 2, so $\text{reciprocal rank} = 1/2 = 0.5$.

If you feel lost, remember this

Precision measures cleanliness, recall measures coverage, and MRR measures how early usefulness begins.

Validation, holdout, and test

Why this chapter exists

You cannot trust an evaluation protocol unless you understand where each decision was made. This chapter explains why training, validation, holdout, and final unlabeled test must be kept conceptually separate.

Why training data alone are not enough

If you choose your model and tune its settings using the same labeled data that taught the model everything, your score will often look too optimistic. You need a protected place to make choices and a cleaner place to check the chosen system later.

Three roles

Split	Has labels?	Main purpose	Wrong use
Training split	Yes	Fit model parameters	Reporting it as the final honest score
Validation split	Yes	Compare settings and choose hyperparameters	Tuning repeatedly on the holdout instead
Holdout split	Yes	One later unbiased check after selection	Using it again and again to tune settings
Final unlabeled test	No visible labels	Submission or deployment-style output	Pretending it can support offline tuning

What leakage is

Leakage means information from the evaluation target influences training or tuning in a way that makes the performance look better than it really is.

Tiny worked example

Suppose you try ten values of `top_k` and pick the best one by looking at the holdout set each time. That holdout set is no longer a clean later check. It has become part of tuning, so the estimate is biased.

Why the final unlabeled test is different

In Kaggle-style setups, the public test queries may have no visible relevance labels. That means they can be used to generate a submission, but not to compute honest offline metrics during development.

Tiny worked example

Project note. In the retrieval-engine dataset used in this project, the files are `queries_train.json`, `qgts_train.json`, and unlabeled `queries_test.json`. That means the clean default protocol is: split the labeled training queries into train and validation, then keep `queries_test.json` only for final submission. Calling an extra internal slice of `queries_train.json` a second “test” split is misleading, because the dataset already provides the real final test file. In addition, `qrels` parsing should be checked explicitly: if the parsed relevance sets are empty or do not line up with the labeled queries, the experiment should stop with an error rather than silently report false zero metrics.

Common mistake

Project audit note. Leakage is not only about using the wrong split. It can also happen through label-revealing metadata. In this project, the query `tags` field was later audited and found to be category-pure in the labeled query set. That means a query classifier using `title + text + tags` can achieve a suspiciously perfect score without any train/validation contamination bug. The correct interpretation is methodological target leakage through metadata, so the brief-aligned `title + text` classifier should remain the main model and the tag-aware version should stay an ablation only.

Formal version

The split logic can be summarized as

fit on train → choose on validation
→ optionally check once on holdout → submit on unlabeled test

The formula is not mathematical; it is a protocol. The order matters.

Common mistake

Retraining after tuning can be valid for final submission generation, but reporting a new score on the same data used in that retraining as if it were an honest evaluation is not valid.

Self-check

1. Why is validation needed even when you already have training labels?
2. What is leakage?
3. Why can an unlabeled competition test set not be used for offline metric comparison?

Short answers

1. Because you need a separate labeled subset for choosing models and hyperparameters honestly.
2. Leakage is unintended flow of evaluation information into training or tuning, making scores too optimistic.
3. Because the relevance labels are not available, so metrics such as recall, precision, or MRR cannot be computed offline.

If you feel lost, remember this

A credible retrieval system is not only a good model. It is also a clean experimental story that explains where each decision came from.

Hyperparameters and tuning

Why this chapter exists

Once the basic protocol is clear, the next question is: which settings should be chosen, and how should they be tuned without breaking the protocol?

What a hyperparameter is

A hyperparameter is a setting chosen by the experimenter rather than learned directly as a model parameter during fitting. In retrieval systems, hyperparameters often control depth, bonus strength, batching, or model selection.

Examples in retrieval pipelines

- `top_k`: how many first-stage candidates to keep,
- `top_m`: how many candidates to rerank with an expensive model,
- category bonus strength λ ,
- embedding batch size,
- choice of retriever, tokenizer, or query representation.

Why tuning is not training

Training changes model parameters using a fixed setup. Tuning compares setups. If you change `top_k` from 500 to 5,000, you did not retrain the meaning of the model itself; you changed the surrounding system design.

Tiny worked example

Suppose you compare three values of `top_k`: 100, 1,000, and 5,000. Increasing `top_k` usually improves recall because more candidates survive. But it may also add noise and raise computational cost. Validation is where you decide which trade-off is best.

Formal version

A simple tuning loop conceptually looks like this:

```
for top_k in candidate_top_ks:
    for top_m in candidate_top_ms:
        for bonus in candidate_bonuses:
            evaluate_on_validation(...)
choose_best_validation_setting()
```

The important point is not the syntax. The important point is that the loop runs on validation, not on holdout.

Common mistake

If you repeatedly tune hyperparameters on the holdout split, that split quietly turns into a second validation set and loses its value as a clean later check.

Self-check

1. Name one hyperparameter that affects retrieval depth.
2. Why can increasing `top_m` improve quality but also increase cost?
3. Where should hyperparameters be chosen: training, validation, or holdout?

Short answers

1. `top_k` or `top_m`.
2. Because reranking more candidates gives the model more chances to fix ordering, but requires more pairwise scoring.
3. They should be chosen on validation.

If you feel lost, remember this

Training learns parameters. Tuning chooses the outer settings of the system. Honest tuning happens on validation, not on holdout.

Part VIII

Engineering and notebook understanding

Why engineering matters

Why this chapter exists

A theoretically good retrieval pipeline can still be unusable if it is too slow or too memory-hungry. Engineering is not decoration around the model; it is part of making the system practical.

Caching

If document embeddings or lexical indexes are expensive to build, recomputing them every run wastes time. **Disk caching** stores artifacts between sessions. **In-memory caching** stores them during one live session to avoid repeated loading.

Embedding reuse

Dense retrieval is a perfect example of reusable computation. The document embeddings do not change when the query changes, so they should usually be computed once and reused.

Chunking

When a matrix of scores becomes too large to fit comfortably in memory, you can split the work into chunks. For example, score 32 queries at a time against the corpus instead of scoring all queries at once.

Partial sorting

If you only need the best k documents, a full exact sort over every score can be wasteful. Partial-selection methods can find the top region more cheaply and then sort only that region.

Tiny worked example

Need the best 5,000 documents out of 216,041? It is usually cheaper to find the top region first and fully sort only that region than to fully sort the whole array.

Runtime versus correctness

Engineering changes can alter runtime without changing the mathematical objective. Chunking and caching are examples. But some engineering choices, such as approximate search, may trade a little exactness for speed. Good system design makes those trade-offs explicit.

Common mistake

Do not treat engineering as unacademic. In retrieval, latency, memory footprint, and reproducibility are part of the real system, not afterthoughts.

Self-check

1. Why is caching especially important for dense retrieval?
2. What problem does chunking solve?
3. Why might a full sort be unnecessary?

Short answers

1. Because document embedding computation is expensive and the same embeddings can be reused across many queries or runs.
2. It controls memory usage by breaking a large computation into smaller pieces.
3. Because in retrieval we often need only the best k items, not a perfectly ordered list of every item in the corpus.

If you feel lost, remember this

A retrieval notebook becomes a retrieval system only when its ideas are paired with runtime-aware design.

Reading the notebook as a system

Why this chapter exists

Once the concepts are clear, the notebook stops looking like random Python. It becomes a system with distinct blocks, each block solving a specific problem.

Conceptual blocks

A retrieval notebook usually contains these stages:

1. data loading and schema checks,
2. text construction and preprocessing,
3. representation building for lexical and dense models,
4. first-stage retrieval,
5. optional classification,
6. optional reranking,
7. evaluation and tuning,
8. final export or submission.

Translate variable names into plain English

Typical name	Plain-English meaning
<code>docs_df</code>	the documents table, that is, the searchable corpus
<code>train_queries_df</code>	the labeled training queries
<code>test_queries_df</code>	the unlabeled final queries for submission
<code>ground_truth</code> or <code>qrels</code>	the answer key telling which documents are relevant to which training queries
<code>base_results</code>	the retrieved candidates before reranking
<code>reranked_results</code>	the same candidates after a second-stage reorder
<code>best_experiment</code>	the validation-selected configuration

Read code by asking four questions

Whenever one cell feels dense, ask:

1. What are the inputs?
2. What new object is being created?
3. Is this a fitting step, a retrieval step, a reranking step, or an evaluation step?
4. Which conceptual stage of the full pipeline does it belong to?

Formal version

A minimal notebook-level pseudocode view is

```
load_data()
build_text_fields()
fit_or_load_retrievers()
retrieve_candidates()
fit_or_load_classifier()
apply_bonus_or_reranker()
evaluate_on_validation()
select_best_setting()
run_final_submission()
```

Read it as pipeline logic, not as code to memorize line by line.

Common mistake

Students often get stuck on variable names. Names matter less than roles. Ask whether the object is data, a model, a score list, or a chosen configuration.

Self-check

1. If you see a variable named `base_results`, what should you suspect it contains?
2. What is the purpose of `best_experiment`?
3. Why should notebook reading begin from pipeline roles rather than isolated lines?

Short answers

1. It likely contains the first-stage candidate rankings before reranking.
2. It stores the configuration that performed best on validation.
3. Because the notebook is a system; understanding the role of each block makes the code interpretable.

If you feel lost, remember this

A notebook becomes readable when each cell is translated into a pipeline responsibility.

One-query journey through the whole pipeline

Why this chapter exists

Many students understand the pieces separately but still do not “see the movie.” This chapter follows one query through the entire pipeline from raw text to evaluation.

Stage 1: raw query

Suppose the user writes:

android application crashes immediately after startup

At this moment, the query is just raw text.

Stage 2: preprocessing

The system normalizes the text, for example by lowercasing and standardizing separators. The query becomes a cleaner representation suitable for the chosen retrieval models.

Stage 3: first-stage retrieval

A lexical retriever or a dense bi-encoder retriever searches the full corpus and returns a candidate set. Suppose the top candidates are:

d_1 (android startup crash), d_2 (android image loading freeze), d_3 (unix service startup failure)

Stage 4: optional category prediction

A classifier predicts that the query category is likely *android*. This information is stored as a weak hint.

Stage 5: reranking

A cross-encoder reads the query with the top candidates jointly. It may decide that d_1 is clearly the most relevant, d_2 is plausible but less exact, and d_3 is less appropriate despite partial similarity.

Stage 6: category adjustment

If category-aware reranking is used, the *android* candidates may receive a small bonus. That reinforces d_1 and d_2 without necessarily deleting d_3 .

Stage 7: final ranking

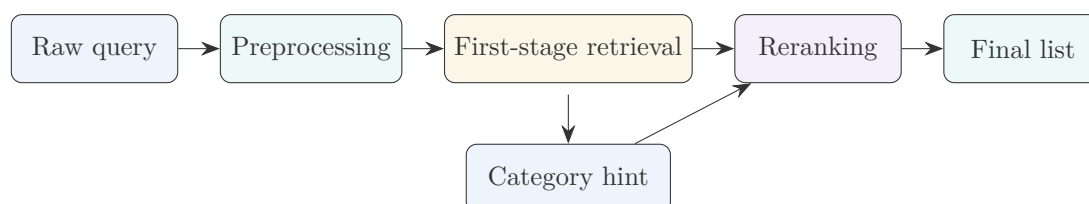
The final list might become

$$[d_1, d_2, d_3, \dots].$$

The user sees the highest-ranked documents first.

Stage 8: evaluation

If this was a validation query and the known relevant set included d_1 , then the pipeline can compute where d_1 appeared, how many relevant documents were recovered, and whether the reranker improved the position.



Common mistake

Notice how each stage adds something different: preprocessing adds consistency, retrieval adds coverage, reranking adds precision, and category adjustment adds a weak topical prior.

Self-check

1. Which stage first narrows the huge corpus to a manageable shortlist?
2. Which stage is most likely to move one candidate above another inside that shortlist?
3. Why is category adjustment described as weak rather than decisive?

Short answers

1. First-stage retrieval.
2. Reranking.
3. Because category is only a supporting signal; exact relevance should still dominate the decision.

If you feel lost, remember this

A useful mental image is: preprocess for consistency, retrieve for coverage, rerank for sharper judgment, and then evaluate the final ordering.

Full pipeline mental model

Why this chapter exists

This final chapter integrates the full learning path. Its goal is not to introduce new terminology, but to connect every major component into one coherent system-level understanding.

End-to-end summary

A retrieval pipeline begins with text representation. Raw documents and queries are normalized and turned into forms that models can use. Sparse lexical models such as TF-IDF and BM25 measure weighted token overlap. Dense semantic models such as bi-encoders map texts into an embedding space where paraphrases can still be close. Because first-stage retrieval must remain scalable, it often serves as a candidate generator. A second-stage reranker, especially a cross-encoder, can then inspect the best candidates more carefully. A classifier may provide a soft topical hint. Evaluation separates training, validation, optional holdout checking, and final unlabeled submission. Engineering makes the whole process practical through caching, chunking, and efficient top- k selection.

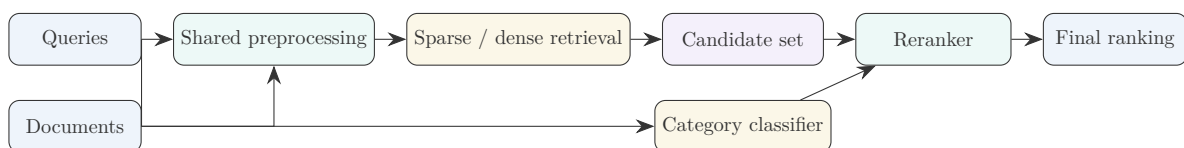
What each stage adds

Stage	Main contribution
Preprocessing	Consistent text representation
Sparse lexical retrieval	Strong exact-token matching and transparency
Dense bi-encoder retrieval	Better semantic coverage and paraphrase recovery
Candidate set	Makes later expensive judgment possible
Cross-encoder reranking	Better local precision among plausible candidates
Category classifier	Weak topical prior or soft bonus
Validation and optional holdout protocol	Honest selection and credible reporting
Engineering	Feasible runtime and reproducible experimentation

When each method helps most

Use lexical methods when exact terms, codes, flags, and commands matter. Use dense retrieval when wording varies and paraphrase mismatch is common. Use reranking when the first stage is good at finding plausible candidates but not perfect at ordering them. Use category information when topical boundaries exist and can help guide near-ties.

Final integrated diagram



Final advice for studying

On your next pass through the notebook, do not ask “what does this line of code do?” first. Ask “which part of the pipeline is this line serving?” That shift in perspective is what turns the notebook from a wall of implementation details into a system you can reason about.

Self-check

1. In one sentence, what is the main job of first-stage retrieval?
2. In one sentence, what is the main job of reranking?
3. Why is evaluation protocol part of the pipeline, not only an external report?

Short answers

1. First-stage retrieval must search the large corpus efficiently and keep a candidate set with good coverage.
2. Reranking must reorder those candidates more precisely using a stronger but more expensive relevance signal.
3. Because the meaning of every reported result depends on where the decisions were made and how tuning was controlled.

If you feel lost, remember this

The full pipeline is a division of labor: represent text well, retrieve broadly, rerank carefully, use category only as support, evaluate honestly, and engineer the system so it can actually run.

Part I glossary

Term	Meaning in plain language
Data	Recorded examples or observations that a system can use.
Feature	A measurable description of an example.
Label	The outcome or target we want to predict.
Model	A mathematical rule that maps input to output.
Training	The process of fitting a model from past examples.
Prediction	Using a fitted model on a new example.
Token	A unit of text used by the system, often a word or word-piece.
Normalization	Making text more consistent before later steps.
Preprocessing	The broader text-preparation stage.
Vector	An ordered list of numbers.
Matrix	A table of numbers, often many vectors stacked together.
Sparse	Mostly zero.
Dense	Many non-zero entries carrying signal together.

Part II glossary

Term	Meaning in plain language
Corpus	The full searchable document collection.
Query	The search request or problem statement.
Document	One searchable item in the corpus.
Relevance	Whether a document truly helps answer the query.
Ranking	The ordered list returned by the system.
Top-k	The first k highest-ranked results.
Lexical matching	Matching based mainly on token overlap.
Semantic matching	Matching based on closeness in meaning.
Vocabulary mismatch	A relevant query-document pair using different words.

Part III glossary

Term	Meaning in plain language
Bag of words	A text view that keeps token counts and mostly ignores order.
TF	Term frequency inside one document.
DF	Number of documents containing a term.
IDF	Inverse document frequency; a weight that grows for rarer terms.
TF-IDF	A sparse weighting scheme combining within-document frequency and corpus rarity.
BM25	A lexical ranking function with term-frequency saturation and length normalization.
BM25+	A refinement of BM25 with a positive offset to reduce over-penalization.

Part IV glossary

Term	Meaning in plain language
Embedding	A dense learned vector representation of text.
Bi-encoder	A model that encodes query and document separately.
Dense retrieval	Retrieval based on learned dense vectors and similarity search.
Similarity search	Finding items whose vectors are close to a query vector.
Recall-oriented candidate generator	A first-stage retriever that aims to keep many relevant items in the shortlist.

Part V glossary

Term	Meaning in plain language
Reranker	A second-stage model that reorders a candidate set.
Candidate set	The shortlist of documents passed from retrieval to later stages.
Cross-encoder	A model that reads the query-document pair jointly and outputs one score.
Positive pair	A query paired with a relevant document.
Negative pair	A query paired with a non-relevant document.
Hard negative	A plausible but still non-relevant document, often mined from retrieval mistakes.

Part VI glossary

Term	Meaning in plain language
Classification	Predicting a category label for one input example.
Category prior	A weak expectation about which topical region is more likely.
Soft bonus	A small score increase rather than a hard removal.
Hard filtering	Removing candidates that fail a category condition.
Linear classifier	A model using weighted sums of input features to score classes.
LinearSVC	A classical linear support-vector classifier often used with TF-IDF features.

Part VII glossary

Term	Meaning in plain language
Precision	Fraction of returned results that are relevant.
Recall	Fraction of all relevant results that were recovered.
F1	A balance of precision and recall.
Reciprocal rank	Inverse of the rank of the first relevant result.
MRR	Mean reciprocal rank across many queries.
Validation split	Held-out labeled data used for model and hyperparameter selection.
Holdout split	A later protected labeled check after selection.
Leakage	Unwanted flow of evaluation information into training or tuning.
Hyperparameter	A system setting chosen by the experimenter rather than directly learned by fitting.

Part VIII glossary

Term	Meaning in plain language
Caching	Saving reusable artifacts instead of recomputing them.
Chunking	Breaking a large computation into smaller pieces.
Partial sorting	Finding only the top region instead of fully sorting everything.
Pipeline	The sequence of stages that turn raw inputs into final outputs.

Final cheat sheet

Why this chapter exists

This cheat sheet is not a substitute for the full guide. It is a memory aid for your second or third pass.

Core distinctions

- Retrieval ranks documents for a query; classification predicts a category for one input.
- Sparse representations emphasize lexical overlap; dense representations emphasize learned semantic closeness.
- Bi-encoders scale to large corpora; cross-encoders usually rerank shortlists.
- Validation chooses settings; an optional holdout can check the chosen system; unlabeled test is for final output, not offline tuning.
- Metadata such as tags can be useful, but if they are nearly identical to the class labels, they should be treated as a label-revealing ablation rather than as the main classifier input.

Formula reminders

$$\text{tfidf}(t, d) = \text{tf}(t, d) \cdot \text{idf}(t)$$

$$\text{idf}(t) = \log\left(\frac{N}{df(t)}\right)$$

$$\cos(\theta) = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\| \|\mathbf{d}\|}$$

$$\text{Precision@K} = \frac{|R_q \cap \hat{R}_{q,K}|}{K}$$

$$\text{Recall@K} = \frac{|R_q \cap \hat{R}_{q,K}|}{|R_q|}$$

$$\text{RR}(q) = \frac{1}{r_q}$$

Pipeline memory line

If you feel lost, remember this

Represent text well, retrieve broadly, rerank carefully, use classification softly, evaluate honestly, and engineer for runtime.

Master glossary

Term	Meaning in plain language
Bag of words	Count-based text view that mostly ignores order.
Batch size	How many examples are processed together in one step.
Bi-encoder	Separate query/document encoder architecture for scalable semantic retrieval.
BM25	Lexical ranking model with saturation and length normalization.
BM25+	BM25 with a positive offset to reduce over-penalization.
Candidate set	Shortlist passed to later stages.
Category bonus	Small score increase when category agreement occurs.
Classification	Predicting a discrete label for one example.
Corpus	Complete searchable document collection.
Cosine similarity	Alignment-based similarity between vectors.
Cross-encoder	Pairwise joint-reading reranker.
Dense vector	Representation with many non-zero values.
Embedding	Dense learned vector for text.
Feature	Input description used by a system.
Hard negative	Plausible but wrong document used in training.
Holdout	Protected later evaluation split.
Hyperparameter	Experimenter-chosen setting such as top-k or bonus strength.
IDF	Inverse document frequency.

Master glossary (continued)

Term	Meaning in plain language
Leakage	Improper use of evaluation information during training or tuning.
MRR	Mean reciprocal rank.
Normalization	Making text more consistent.
Precision	How clean returned results are.
Query	The search request.
Recall	How much relevant material was recovered.
Retrieval	Ranking documents for a query.
Reranker	Second-stage model that reorders a candidate set.
Sparse vector	Vector with mostly zero entries.
TF	Term frequency.
TF-IDF	Sparse lexical weighting scheme.
Token	Text unit used by the system.
top-k	Number of candidates retained after a stage.
top-m	Number of candidates passed to an expensive reranker.
Validation	Split used to choose settings honestly.
Vocabulary mismatch	Relevant texts using different words.
Vector	Ordered list of numbers.